

Supplementary Model Information

S1. General information

The seismic design intensity is 8 (0.2g), and the site characteristic period is 0.35 s. The floor and roof dead and live loads are 6.0 and 2.0 kN/m², respectively.

Table S1. General information on the frame models

Frame	Concrete	No. of spans	Span length (m)	Story height, first/typical (m)	Column section (m × m)	Beam section (m × m)	Fundamental period, T ₁ (s)	
							RCF	RCFAPDYBRB
5-IV	C30	3	6	3.6/3.3	0.50 × 0.50	0.50 × 0.25	0.9175	0.7756
5-SD	C30	3	6	3.6/3.3	0.50 × 0.50	0.50 × 0.25	0.9175	0.7719
5-V	C30	3	6	3.6/3.3	0.50 × 0.50	0.50 × 0.25	0.9175	0.7782
7-IV	C30	3	6	3.6/3.3	0.55 × 0.55	0.50 × 0.25	1.2434	1.0012
7-SD	C30	3	5	3.9/3.6	0.55 × 0.55	0.50 × 0.25	1.2434	1.0395
7-V	C30	3	6	3.6/3.3	0.55 × 0.55	0.50 × 0.25	1.2434	1.0071
9-IV	C40	3	6	3.6/3.3	0.55 × 0.55	0.50 × 0.30	1.2488	1.0754
9-SD	C40	3	5	3.9/3.6	0.55 × 0.55	0.50 × 0.30	1.2488	1.1980
9-V	C40	3	6	3.6/3.3	0.55 × 0.55	0.50 × 0.30	1.2488	1.0796
11-IV	C40	3	6	3.6/3.3	0.60 × 0.60	0.50 × 0.30	1.4936	1.2803
11-SD	C40	3	6	3.6/3.3	0.60 × 0.60	0.50 × 0.30	1.4936	1.2717
11-V	C40	3	6	3.6/3.3	0.60 × 0.60	0.50 × 0.30	1.4936	1.2849

Note: The two story-height values denote the first-story and typical-story heights, respectively.

S2. Frame configurations and reinforcement

Table S2. Beam and column reinforcement ratios of the 5-story frame with inverted-V bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.57	0.30	0.30	0.70	0.41	0.60	0.30	0.53
2	0.53	0.30	0.30	0.65	0.41	0.61	0.31	0.74
3	0.50	0.30	0.30	0.61	0.41	0.62	0.40	0.83
4	0.50	0.30	0.30	0.58	0.40	0.62	0.41	0.78
5	0.50	0.30	0.30	0.55	0.40	0.62	0.30	0.53

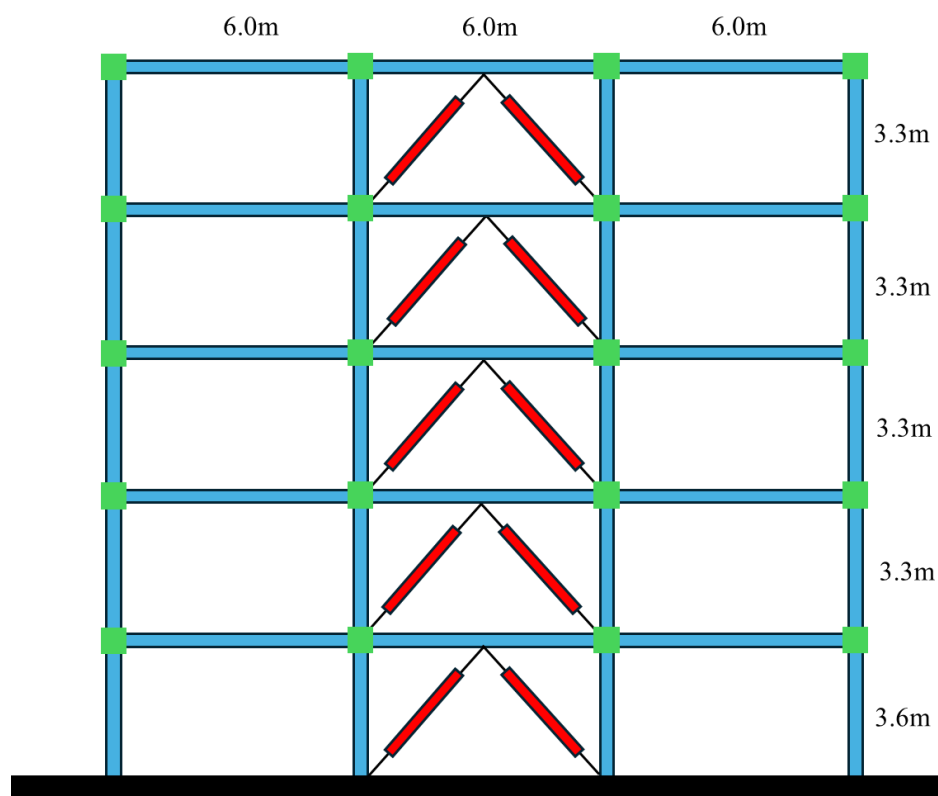


Fig. S1. Elevation of the 5-story frame with inverted-V bracing

Table S3. Beam and column reinforcement ratios of the 5-story frame with single-diagonal bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.57	0.30	0.30	0.70	0.41	0.60	0.30	0.53
2	0.53	0.30	0.30	0.65	0.41	0.61	0.31	0.74
3	0.50	0.30	0.30	0.61	0.41	0.62	0.40	0.83
4	0.50	0.30	0.30	0.58	0.40	0.62	0.41	0.78
5	0.50	0.30	0.30	0.55	0.40	0.62	0.30	0.53

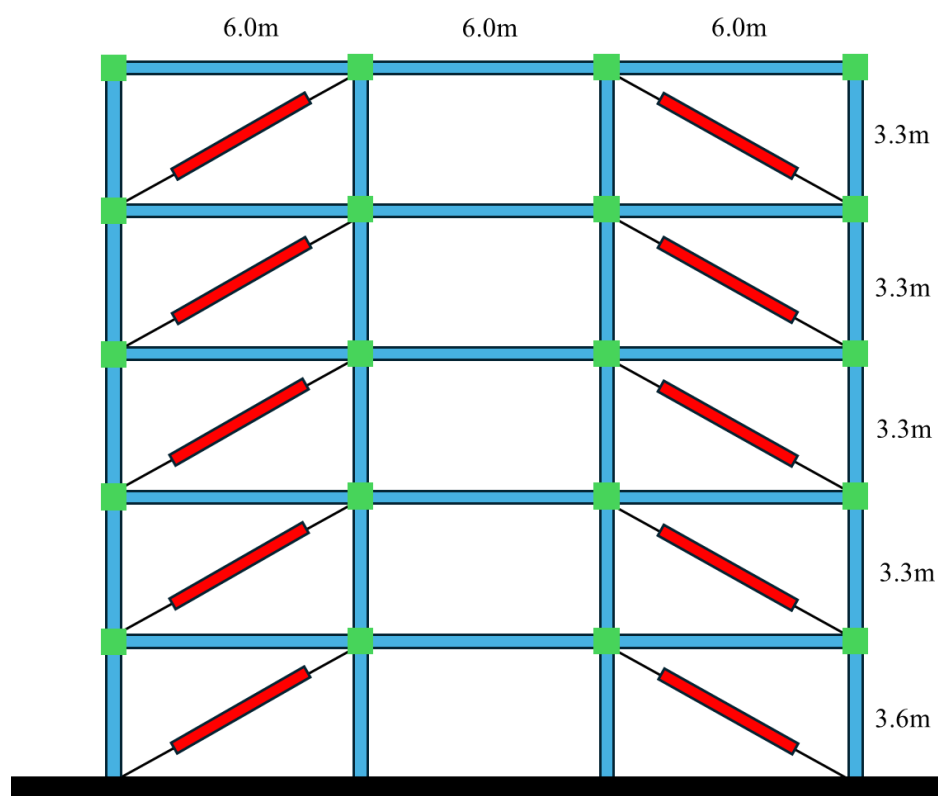


Fig. S2. Elevation of the 5-story frame with single-diagonal bracing

Table S4. Beam and column reinforcement ratios of the 5-story frame with V-shaped bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.56	0.30	0.30	0.73	0.41	0.64	0.30	0.54
2	0.53	0.30	0.30	0.67	0.41	0.64	0.31	0.73
3	0.50	0.30	0.30	0.61	0.40	0.65	0.40	0.79
4	0.50	0.30	0.30	0.56	0.40	0.64	0.41	0.72
5	0.50	0.30	0.30	0.50	0.30	0.56	0.30	0.46

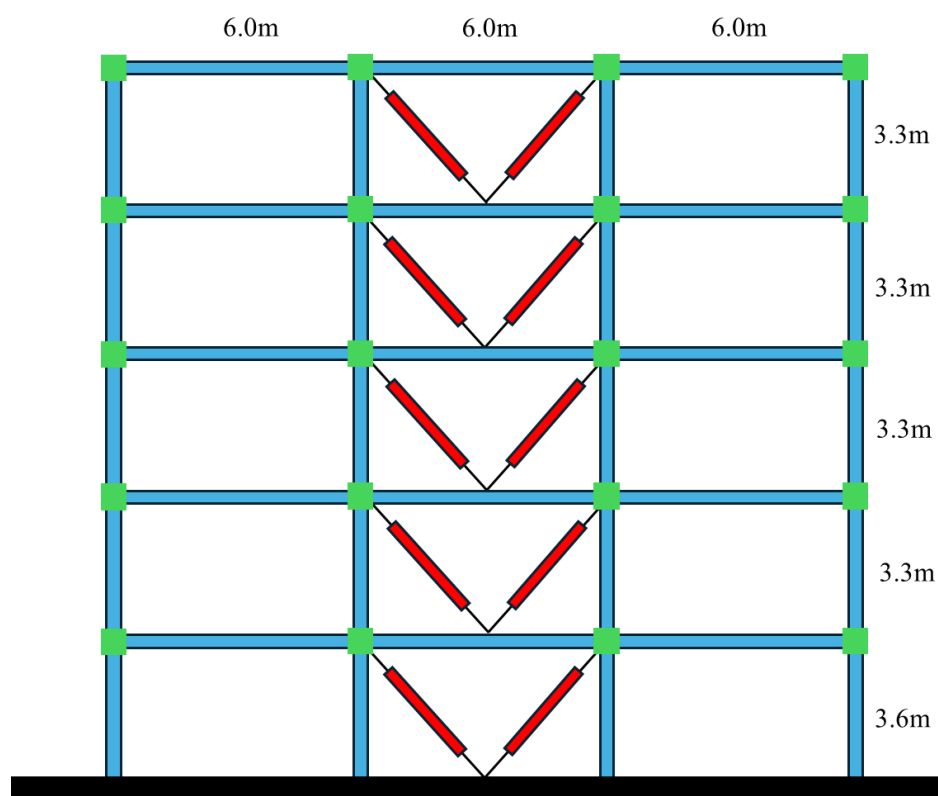


Fig. S3. Elevation of the 5-story frame with V-shaped bracing

Table S5. Beam and column reinforcement ratios of the 7-story frame with inverted-V bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.67	0.34	0.34	0.83	0.44	0.58	0.30	0.37
2	0.65	0.33	0.33	0.80	0.42	0.57	0.30	0.55
3	0.61	0.31	0.31	0.75	0.41	0.58	0.30	0.66
4	0.56	0.30	0.30	0.68	0.41	0.60	0.30	0.70
5	0.50	0.30	0.30	0.60	0.40	0.61	0.32	0.68
6	0.50	0.30	0.30	0.58	0.40	0.61	0.31	0.59
7	0.50	0.30	0.30	0.55	0.40	0.61	0.30	0.38

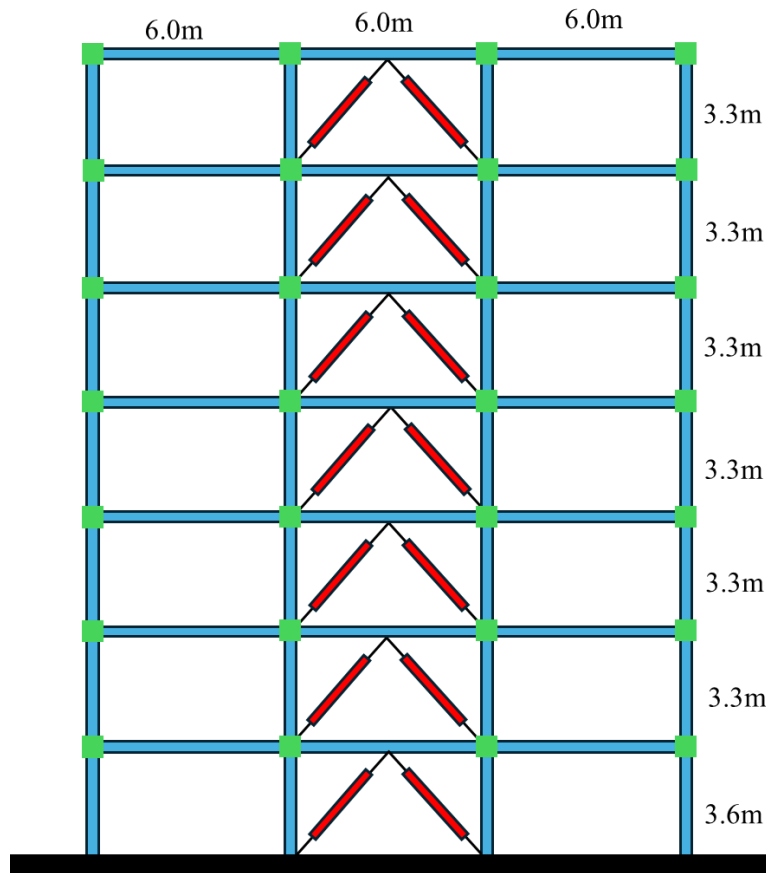


Fig. S4. Elevation of the 7-story frame with inverted-V bracing

Table S6. Beam and column reinforcement ratios of the 7-story frame with single-diagonal bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.45	0.42	0.42	0.52	0.40	0.40	0.37	0.37
2	0.43	0.42	0.42	0.50	0.40	0.40	0.37	0.48
3	0.41	0.41	0.41	0.47	0.40	0.40	0.37	0.60
4	0.41	0.41	0.41	0.43	0.40	0.40	0.37	0.65
5	0.41	0.41	0.41	0.40	0.40	0.40	0.37	0.65
6	0.41	0.41	0.41	0.40	0.40	0.40	0.37	0.56
7	0.40	0.40	0.40	0.40	0.40	0.40	0.37	0.37

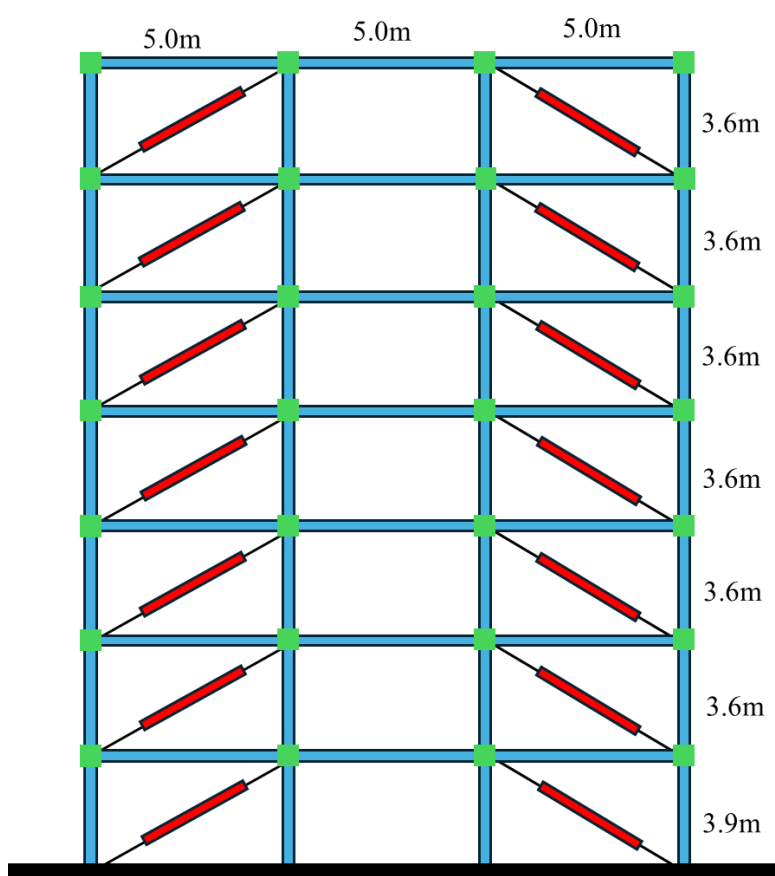


Fig. S5. Elevation of the 7-story frame with single-diagonal bracing

Table S7. Beam and column reinforcement ratios of the 7-story frame with V-shaped bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.67	0.34	0.34	0.83	0.44	0.58	0.30	0.37
2	0.65	0.33	0.33	0.80	0.42	0.57	0.30	0.55
3	0.61	0.31	0.31	0.75	0.41	0.58	0.30	0.66
4	0.56	0.30	0.30	0.68	0.41	0.60	0.30	0.70
5	0.50	0.30	0.30	0.60	0.40	0.61	0.32	0.68
6	0.50	0.30	0.30	0.58	0.40	0.61	0.31	0.59
7	0.50	0.30	0.30	0.55	0.40	0.61	0.30	0.38

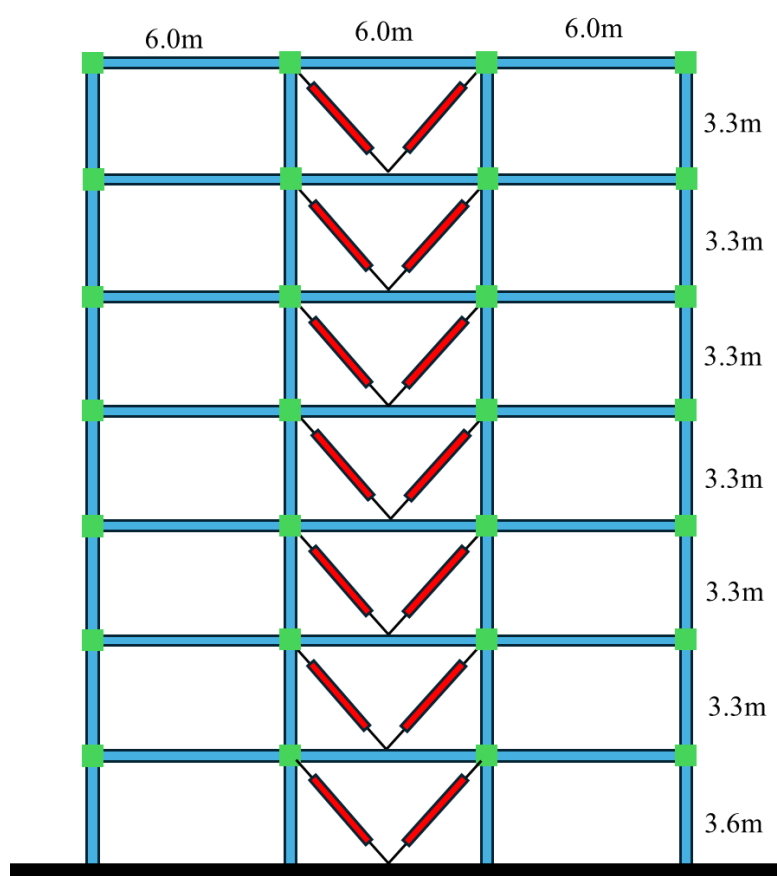


Fig. S6. Elevation of the 7-story frame with V-shaped bracing

Table S8. Beam and column reinforcement ratios of the 9-story frame with inverted-V bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.51	0.53	0.53	0.71	0.53	0.64	0.37	0.57
2	0.50	0.53	0.53	0.69	0.53	0.64	0.37	0.84
3	0.48	0.53	0.53	0.65	0.53	0.64	0.37	1.02
4	0.46	0.53	0.53	0.61	0.53	0.65	0.45	1.14
5	0.44	0.53	0.53	0.55	0.53	0.65	0.50	1.20
6	0.44	0.53	0.53	0.49	0.53	0.67	0.51	1.18
7	0.44	0.53	0.53	0.47	0.53	0.67	0.48	1.08
8	0.44	0.53	0.53	0.44	0.53	0.67	0.40	0.86
9	0.44	0.53	0.53	0.44	0.53	0.65	0.37	0.53

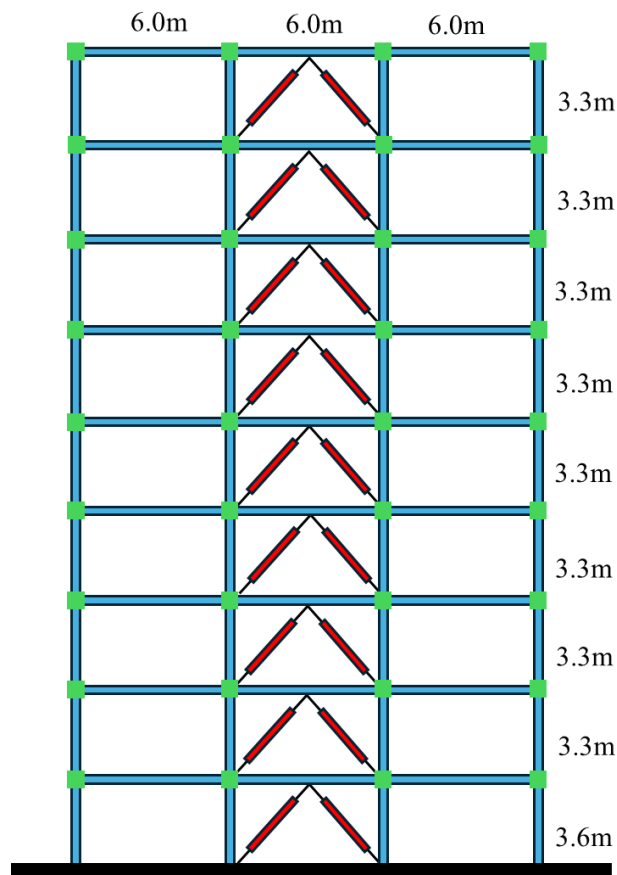


Fig. S7. Elevation of the 9-story frame with inverted-V bracing

Table S9. Beam and column reinforcement ratios of the 9-story frame with single-diagonal bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.46	0.55	0.55	0.44	0.53	0.53	0.49	0.59
2	0.46	0.55	0.55	0.44	0.53	0.53	0.60	0.90
3	0.46	0.55	0.55	0.44	0.53	0.53	0.67	1.13
4	0.46	0.55	0.55	0.44	0.53	0.53	0.71	1.27
5	0.45	0.54	0.54	0.44	0.53	0.53	0.71	1.33
6	0.45	0.54	0.54	0.44	0.53	0.53	0.67	1.30
7	0.45	0.54	0.54	0.44	0.53	0.53	0.59	1.17
8	0.45	0.53	0.53	0.44	0.53	0.53	0.45	0.93
9	0.44	0.53	0.53	0.44	0.53	0.53	0.37	0.56

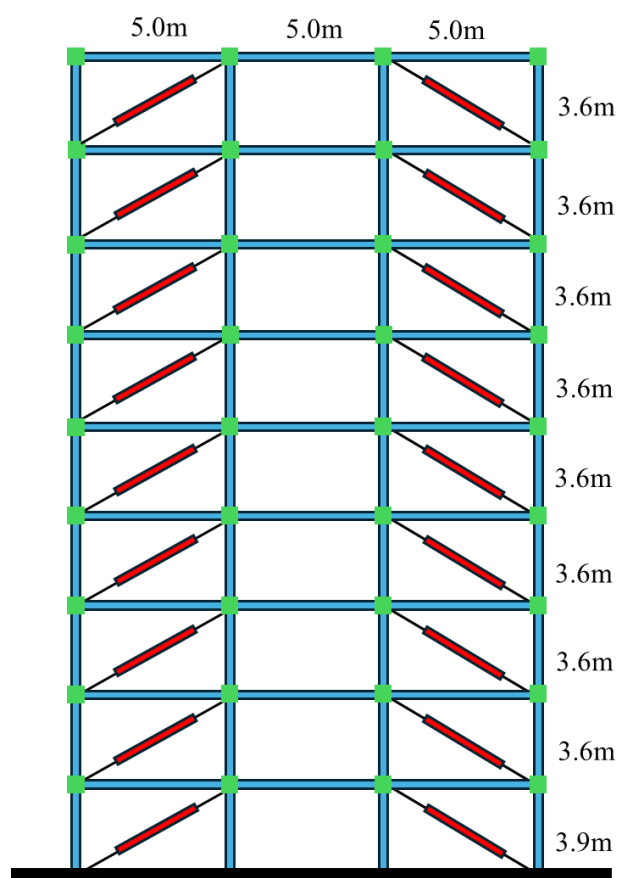


Fig. S8. Elevation of the 9-story frame with single-diagonal bracing

Table S10. Beam and column reinforcement ratios of the 9-story frame with V-shaped bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.51	0.53	0.53	0.71	0.53	0.64	0.37	0.57
2	0.50	0.53	0.53	0.69	0.53	0.64	0.37	0.84
3	0.48	0.53	0.53	0.65	0.53	0.64	0.37	1.02
4	0.46	0.53	0.53	0.61	0.53	0.65	0.45	1.14
5	0.44	0.53	0.53	0.55	0.53	0.65	0.50	1.20
6	0.44	0.53	0.53	0.49	0.53	0.67	0.51	1.18
7	0.44	0.53	0.53	0.47	0.53	0.67	0.48	1.08
8	0.44	0.53	0.53	0.44	0.53	0.67	0.40	0.86
9	0.44	0.53	0.53	0.44	0.53	0.65	0.37	0.53

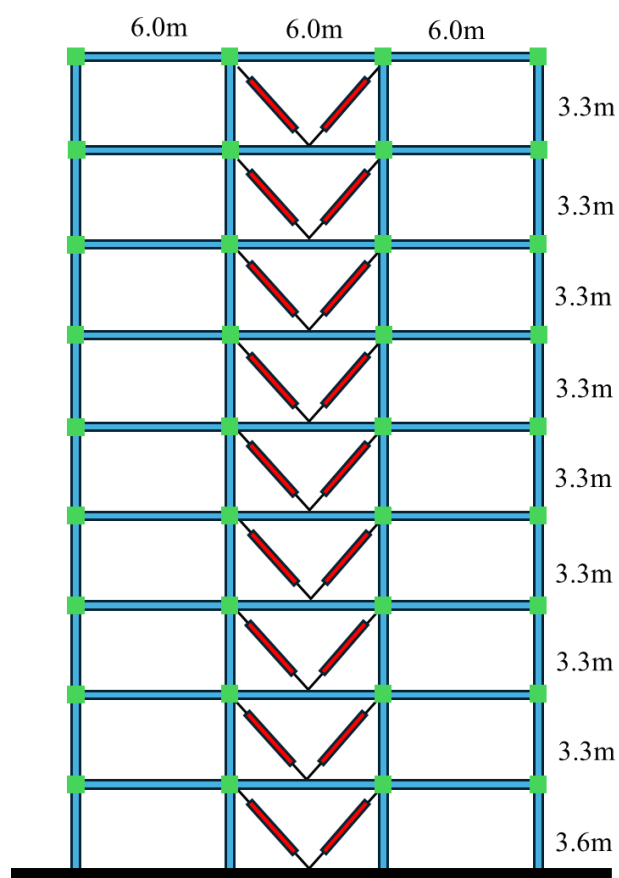


Fig. S9. Elevation of the 9-story frame with V-shaped bracing

Table S11. Beam and column reinforcement ratios of the 11-story frame with inverted-V bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.56	0.53	0.53	0.74	0.53	0.59	0.37	0.37
2	0.56	0.53	0.53	0.72	0.53	0.59	0.37	0.59
3	0.54	0.53	0.53	0.70	0.53	0.59	0.37	0.76
4	0.52	0.53	0.53	0.67	0.53	0.60	0.37	0.88
5	0.50	0.53	0.53	0.64	0.53	0.60	0.37	0.95
6	0.47	0.53	0.53	0.59	0.53	0.61	0.41	0.99
7	0.44	0.53	0.53	0.54	0.53	0.62	0.43	0.99
8	0.44	0.53	0.53	0.48	0.53	0.64	0.42	0.94
9	0.44	0.53	0.53	0.46	0.53	0.64	0.37	0.83
10	0.44	0.53	0.53	0.44	0.53	0.65	0.37	0.65
11	0.44	0.53	0.53	0.44	0.53	0.65	0.37	0.39

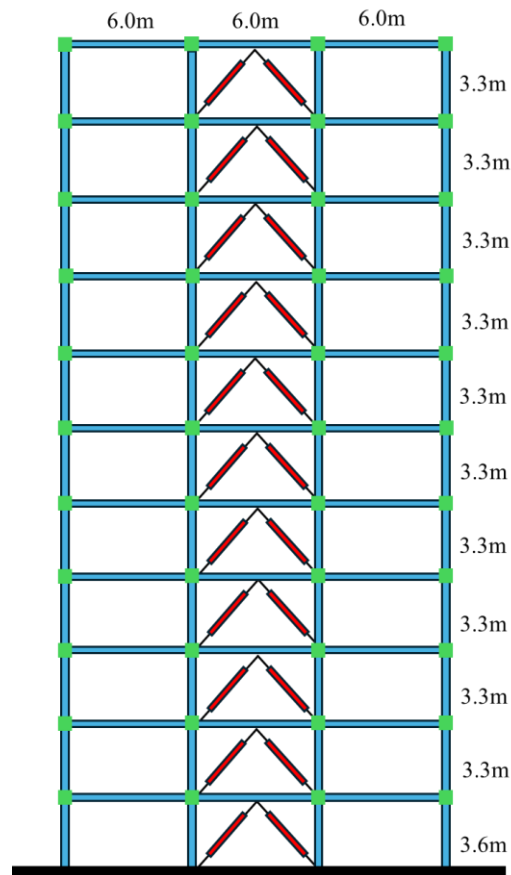


Fig. S10. Elevation of the 11-story frame with inverted-V bracing

Table S12. Beam and column reinforcement ratios of the 11-story frame with single-diagonal bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.56	0.53	0.53	0.74	0.53	0.59	0.37	0.37
2	0.56	0.53	0.53	0.72	0.53	0.59	0.37	0.59
3	0.54	0.53	0.53	0.70	0.53	0.59	0.37	0.76
4	0.52	0.53	0.53	0.67	0.53	0.60	0.37	0.88
5	0.50	0.53	0.53	0.64	0.53	0.60	0.37	0.95
6	0.47	0.53	0.53	0.59	0.53	0.61	0.41	0.99
7	0.44	0.53	0.53	0.54	0.53	0.62	0.43	0.99
8	0.44	0.53	0.53	0.48	0.53	0.64	0.42	0.94
9	0.44	0.53	0.53	0.46	0.53	0.64	0.37	0.83
10	0.44	0.53	0.53	0.44	0.53	0.65	0.37	0.65
11	0.44	0.53	0.53	0.44	0.53	0.65	0.37	0.39

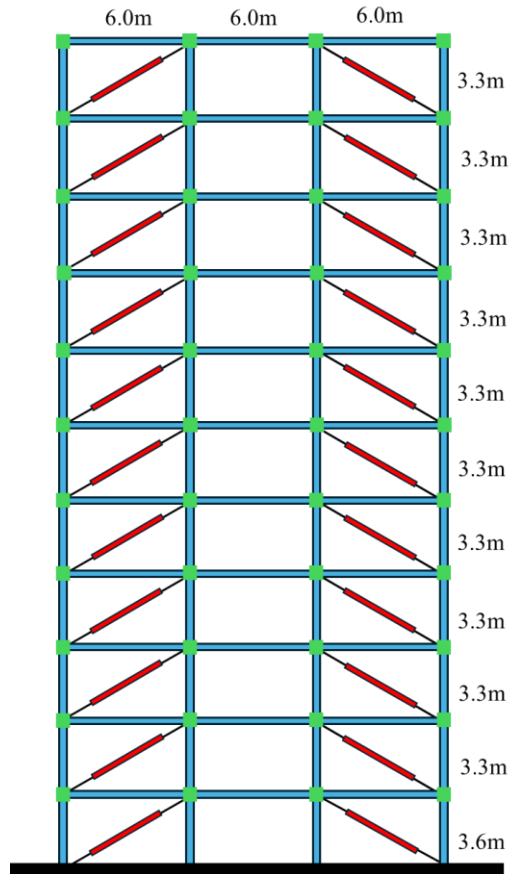


Fig. S11. Elevation of the 11-story frame with single-diagonal bracing

Table S13. Beam and column reinforcement ratios of the 11-story frame with V-shaped bracing

Story	Beam reinforcement ratio (%)						Column reinforcement ratio (%)	
	Exterior span			Interior span			Exterior column	Interior column
	Top	Bottom at beam end	Bottom at midspan	Top	Bottom at beam end	Bottom at midspan	Each face	Each face
1	0.56	0.53	0.53	0.78	0.53	0.62	0.37	0.42
2	0.56	0.53	0.53	0.77	0.53	0.62	0.37	0.64
3	0.54	0.53	0.53	0.74	0.53	0.62	0.37	0.80
4	0.52	0.53	0.53	0.71	0.53	0.62	0.37	0.91
5	0.50	0.53	0.53	0.67	0.53	0.63	0.37	0.98
6	0.47	0.53	0.53	0.62	0.53	0.64	0.41	1.02
7	0.44	0.53	0.53	0.55	0.53	0.65	0.43	1.01
8	0.44	0.53	0.53	0.49	0.53	0.66	0.42	0.96
9	0.44	0.53	0.53	0.46	0.53	0.66	0.38	0.85
10	0.44	0.53	0.53	0.44	0.53	0.66	0.37	0.66
11	0.44	0.53	0.53	0.44	0.53	0.64	0.37	0.40

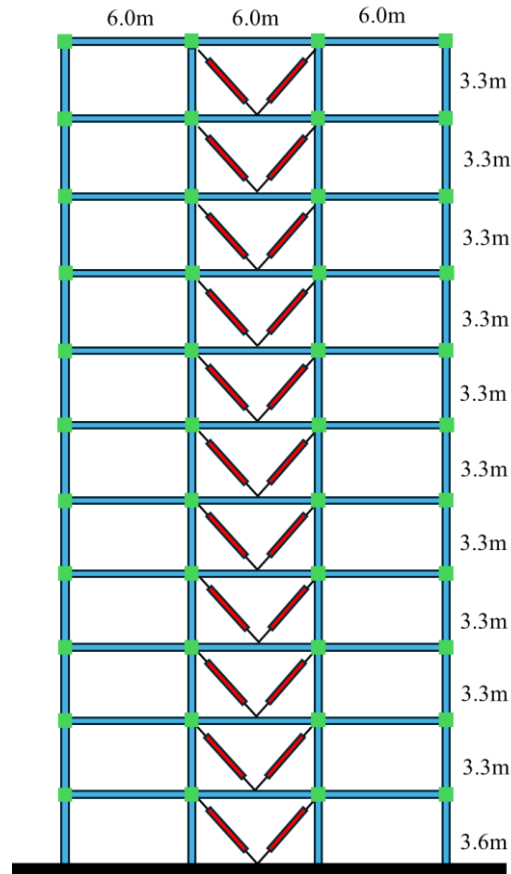


Fig. S12. Elevation of the 11-story frame with V-shaped bracing

S3. APDYBRB parameters

For all APDYBRB cases, $F_{y1} = 0.5F_{y2}$, and $K_1 = K_2$. The MCE values of F_{y2} and Δ_0 are twice their DBE values, whereas the initial stiffnesses remain unchanged.

Table S14. APDYBRB parameters for cases 5-IV-D-0.5 and 5-IV-M-0.5

Story	5-IV-D-0.5			5-IV-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	142680.432	30091.880	4.741	285360.864	30091.880	9.482
2	123649.504	31429.944	3.934	247299.008	31429.944	7.868
3	105741.521	31429.944	3.364	211483.042	31429.944	6.728
4	81303.296	31429.944	2.587	162606.592	31429.944	5.174
5	48564.514	31429.944	1.545	97129.028	31429.944	3.090

Table S15. APDYBRB parameters for cases 5-SD-D-0.5 and 5-SD-M-0.5

Story	5-SD-D-0.5			5-SD-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	142680.432	30091.880	4.741	285360.864	30091.880	9.482
2	123649.504	31429.944	3.934	247299.008	31429.944	7.868
3	105741.521	31429.944	3.364	211483.042	31429.944	6.728
4	81303.296	31429.944	2.587	162606.592	31429.944	5.174
5	48564.514	31429.944	1.545	97129.028	31429.944	3.090

Table S16. APDYBRB parameters for cases 5-V-D-0.5 and 5-V-M-0.5

Story	5-V-D-0.5			5-V-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	142680.432	30091.880	4.741	285360.864	30091.880	9.482
2	123649.504	31429.944	3.934	247299.008	31429.944	7.868
3	105741.521	31429.944	3.364	211483.042	31429.944	6.728
4	81303.296	31429.944	2.587	162606.592	31429.944	5.174
5	48564.514	31429.944	1.545	97129.028	31429.944	3.090

Table S17. APDYBRB parameters for cases 7-IV-D-0.5 and 7-IV-M-0.5

Story	7-IV-D-0.5			7-IV-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	168865.259	35614.366	4.741	337730.518	35614.366	9.482
2	151068.791	37197.992	4.061	302137.582	37197.992	8.122
3	138255.794	37197.992	3.717	276511.588	37197.992	7.434
4	122000.305	37197.992	3.280	244000.610	37197.992	6.560
5	101900.082	37197.992	2.739	203800.164	37197.992	5.478
6	77175.079	37197.992	2.075	154350.158	37197.992	4.150
7	45966.563	37197.992	1.236	91933.126	37197.992	2.472

Table S18. APDYBRB parameters for cases 7-SD-D-0.5 and 7-SD-M-0.5

Story	7-SD-D-0.5			7-SD-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	100104.539	22220.214	4.505	200209.078	22220.214	9.010
2	90836.086	23127.533	3.928	181672.172	23127.533	7.856
3	82634.444	23127.533	3.573	165268.888	23127.533	7.146
4	72493.629	23127.533	3.135	144987.258	23127.533	6.270
5	60182.923	23127.533	2.602	120365.846	23127.533	5.204
6	45253.318	23127.533	1.957	90506.636	23127.533	3.914
7	26627.222	23127.533	1.151	53254.444	23127.533	2.302

Table S19. APDYBRB parameters for cases 7-V-D-0.5 and 7-V-M-0.5

Story	7-V-D-0.5			7-V-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	168865.259	35614.366	4.741	337730.518	35614.366	9.482
2	151068.791	37197.992	4.061	302137.582	37197.992	8.122
3	138255.794	37197.992	3.717	276511.588	37197.992	7.434
4	122000.305	37197.992	3.280	244000.610	37197.992	6.560
5	101900.082	37197.992	2.739	203800.164	37197.992	5.478
6	77175.079	37197.992	2.075	154350.158	37197.992	4.150
7	45966.563	37197.992	1.236	91933.126	37197.992	2.472

Table S20. APDYBRB parameters for cases 9-IV-D-0.5 and 9-IV-M-0.5

Story	9-IV-D-0.5			9-IV-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	183812.588	38766.818	4.741	367625.176	38766.818	9.482
2	166403.349	40490.621	4.110	332806.698	40490.621	8.220
3	155987.156	40490.621	3.852	311974.312	40490.621	7.704
4	143559.050	40490.621	3.545	287118.100	40490.621	7.090
5	128982.489	40490.621	3.185	257964.978	40490.621	6.370
6	112035.430	40490.621	2.767	224070.860	40490.621	5.534
7	92344.955	40490.621	2.281	184689.910	40490.621	4.562
8	69216.658	40490.621	1.709	138433.316	40490.621	3.418
9	41009.787	40490.621	1.013	82019.574	40490.621	2.026

Table S21. APDYBRB parameters for cases 9-SD-D-0.5 and 9-SD-M-0.5

Story	9-SD-D-0.5			9-SD-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	111257.725	24695.888	4.505	222515.450	24695.888	9.010
2	102244.451	25704.296	3.978	204488.902	25704.296	7.956
3	95339.797	25704.296	3.709	190679.594	25704.296	7.418
4	87303.063	25704.296	3.396	174606.126	25704.296	6.792
5	78055.093	25704.296	3.037	156110.186	25704.296	6.074
6	67466.874	25704.296	2.625	134933.748	25704.296	5.250
7	55321.040	25704.296	2.152	110642.080	25704.296	4.304
8	41210.897	25704.296	1.603	82421.794	25704.296	3.206
9	24166.761	25704.296	0.940	48333.522	25704.296	1.880

Table S22. APDYBRB parameters for cases 9-V-D-0.5 and 9-V-M-0.5

Story	9-V-D-0.5			9-V-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	183812.588	38766.818	4.741	367625.176	38766.818	9.482
2	166403.349	40490.621	4.110	332806.698	40490.621	8.220
3	155987.156	40490.621	3.852	311974.312	40490.621	7.704
4	143559.050	40490.621	3.545	287118.100	40490.621	7.090
5	128982.489	40490.621	3.185	257964.978	40490.621	6.370
6	112035.430	40490.621	2.767	224070.860	40490.621	5.534
7	92344.955	40490.621	2.281	184689.910	40490.621	4.562
8	69216.658	40490.621	1.709	138433.316	40490.621	3.418
9	41009.787	40490.621	1.013	82019.574	40490.621	2.026

Table S23. APDYBRB parameters for cases 11-IV-D-0.5 and 11-IV-M-0.5

Story	11-IV-D-0.5			11-IV-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	205334.090	43305.790	4.741	410668.180	43305.790	9.482
2	187363.289	45241.889	4.142	374726.578	45241.889	8.284
3	178026.515	45241.889	3.936	356053.030	45241.889	7.872
4	167315.312	45241.889	3.699	334630.624	45241.889	7.398
5	155164.692	45241.889	3.430	310329.384	45241.889	6.860
6	141476.337	45241.889	3.127	282952.674	45241.889	6.254
7	126103.044	45241.889	2.788	252206.088	45241.889	5.576
8	108818.483	45241.889	2.406	217636.966	45241.889	4.812
9	89251.535	45241.889	1.973	178503.070	45241.889	3.946
10	66714.479	45241.889	1.475	133428.958	45241.889	2.950
11	39564.164	45241.889	0.875	79128.328	45241.889	1.750

Table S24. APDYBRB parameters for cases 11-SD-D-0.5 and 11-SD-M-0.5

Story	11-SD-D-0.5			11-SD-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	205334.090	43305.790	4.741	410668.180	43305.790	9.482
2	187363.289	45241.889	4.142	374726.578	45241.889	8.284
3	178026.515	45241.889	3.936	356053.030	45241.889	7.872
4	167315.312	45241.889	3.699	334630.624	45241.889	7.398
5	155164.692	45241.889	3.430	310329.384	45241.889	6.860
6	141476.337	45241.889	3.127	282952.674	45241.889	6.254
7	126103.044	45241.889	2.788	252206.088	45241.889	5.576
8	108818.483	45241.889	2.406	217636.966	45241.889	4.812
9	89251.535	45241.889	1.973	178503.070	45241.889	3.946
10	66714.479	45241.889	1.475	133428.958	45241.889	2.950
11	39564.164	45241.889	0.875	79128.328	45241.889	1.750

Table S25. APDYBRB parameters for cases 11-V-D-0.5 and 11-V-M-0.5

Story	11-V-D-0.5			11-V-M-0.5		
	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)	Second-stage yield force, F_{y2} (N)	First- and second-stage initial stiffnesses, K_1 and K_2 (N/mm)	Second-stage activation displacement, Δ_0 (mm)
1	205334.090	43305.790	4.741	410668.180	43305.790	9.482
2	187363.289	45241.889	4.142	374726.578	45241.889	8.284
3	178026.515	45241.889	3.936	356053.030	45241.889	7.872
4	167315.312	45241.889	3.699	334630.624	45241.889	7.398
5	155164.692	45241.889	3.430	310329.384	45241.889	6.860
6	141476.337	45241.889	3.127	282952.674	45241.889	6.254
7	126103.044	45241.889	2.788	252206.088	45241.889	5.576
8	108818.483	45241.889	2.406	217636.966	45241.889	4.812
9	89251.535	45241.889	1.973	178503.070	45241.889	3.946
10	66714.479	45241.889	1.475	133428.958	45241.889	2.950
11	39564.164	45241.889	0.875	79128.328	45241.889	1.750